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Image Processing and Computer Vision Project

Comparison between Standard Convolutional Networks LeNet5 and ConvKAN using Grad-CAM

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Chapter 1

Introduction

1.1 Context and Objective

In this project, we analyze the behavior of two convolutional network models using the E-MNIST dataset. This dataset, derived from MNIST, includes both digits and letters (upper- and lowercase) for a total of 62 classes, making it a more complex classification problem than the classic MNIST.

The objective is to compare the classic LeNet5 convolutional network with a convolutional version of the Kolmogorov-Arnold Networks (KAN), known as ConvKAN, in terms of decision interpretability. The use of Grad-CAM will allow us to visually highlight which parts of the image most influence the network decisions.

1.2 Introduction to KAN

Kolmogorov-Arnold Networks (KAN) are based on the Kolmogorov-Arnold theorem, which states that every continuous multivariate function can be represented as a sum of univariate functions. KAN exploit this concept by applying univariate spline functions to the weights instead of using fixed activations at the nodes. In this way, the weights are no longer simple scalars but adaptive spline functions capable of modeling more complex nonlinear relationships than traditional Multi-Layer Perceptrons (MLPs). **(Liu et al., 2024 state: “Kolmogorov-Arnold Networks offer promising alternatives to traditional MLPs” [4]).**

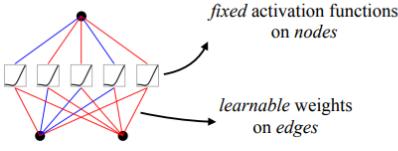
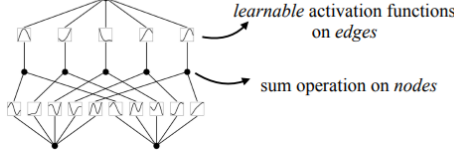
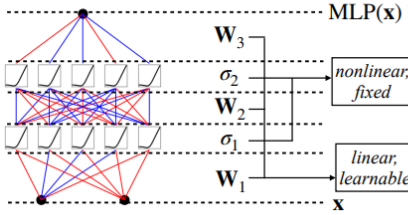
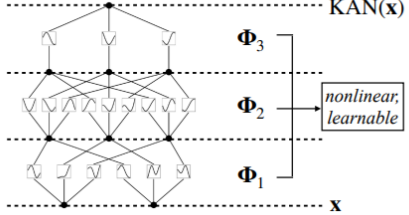
Model	Multi-Layer Perceptron (MLP)	Kolmogorov-Arnold Network (KAN)
Theorem	Universal Approximation Theorem	Kolmogorov-Arnold Representation Theorem
Formula (Shallow)	$f(\mathbf{x}) \approx \sum_{i=1}^{N(\epsilon)} a_i \sigma(\mathbf{w}_i \cdot \mathbf{x} + b_i)$	$f(\mathbf{x}) = \sum_{q=1}^{2n+1} \Phi_q \left(\sum_{p=1}^n \phi_{q,p}(x_p) \right)$
Model (Shallow)	(a) 	(b) 
Formula (Deep)	$\text{MLP}(\mathbf{x}) = (\mathbf{W}_3 \circ \sigma_2 \circ \mathbf{W}_2 \circ \sigma_1 \circ \mathbf{W}_1)(\mathbf{x})$	$\text{KAN}(\mathbf{x}) = (\Phi_3 \circ \Phi_2 \circ \Phi_1)(\mathbf{x})$
Model (Deep)	(c) 	(d) 

Figure 1.1: Comparison between Multi-Layer Perceptron (MLP) and Kolmogorov-Arnold Network (KAN). The diagram highlights the key differences in architecture, formula representation, and learning mechanisms.

1.3 Extension to ConvKAN

ConvKAN extends the principles of KAN to convolution operations. Instead of using rigid linear filters, the filters are represented by learned spline functions during training. Parameters such as `grid_size` and `spline_order` control the complexity of these splines, enabling ConvKAN to capture more complex patterns and potentially offer better interpretability. (Bodner et al., 2024 note: “Convolutional Kolmogorov-Arnold Networks effectively extend KAN principles to convolution operations” [5]).

1.4 Motivation

Interpretability in deep learning models plays a crucial role. Grad-CAM helps visualize the regions of the image deemed important by the model. As Selvaraju et al. (2017) explained, “**Grad-CAM produces visual explanations from deep networks via gradient-based localization**” [3], which is why comparing LeNet5 (a classic architecture) with ConvKAN (an innovative spline-based architecture) is relevant to understand whether the greater expressive power of spline filters translates not only into performance improvements but also into a different internal attention distribution.

Chapter 2

Used Architectures

2.1 LeNet5

LeNet5 follows the following sequence:

- **Convolution:** 1 input, 6 output, kernel 5×5 , padding=2, followed by a ReLU activation.
- **Pooling:** 2×2 window, reducing the size from 28×28 to 14×14 .
- **Convolution:** 6 input, 16 output, kernel 5×5 , without padding, followed by a ReLU activation.
- **Pooling:** 2×2 window, reducing the size from 10×10 to 5×5 .
- **Fully Connected Layers:** A sequence of layers with 256, 120, 84, and 62 neurons, with intermediate ReLU activations.

The filters used are linear, and this network serves as a well-established baseline. (LeCun et al., 1998 explain: “The combination of convolution and pooling layers forms the foundation of effective document recognition” [1]).

2.2 ConvKAN

ConvKAN uses layers called `KAN_Convolutional_Layer` based on splines. Below is the KAN topology used in the LeNet5_KAN (ConvKAN) variant:

- **First conv KAN layer:**
 - `in_channels=1`, `out_channels=6`, kernel 5×5 , `stride=1`, `padding=0`.
 - `grid_size=5`, `spline_order=3`, with ReLU as the base activation.
 - 2×2 Pooling that reduces the size from 28×28 to 12×12 (a 5×5 convolution without padding yields 24×24 , then pooling halves the size).
- **Second conv KAN layer:**
 - `in_channels=6`, `out_channels=16`, kernel 5×5 , `stride=1`, without padding.
 - Same spline settings (`grid_size=5`, `spline_order=3`).

- 2×2 Pooling that reduces the size from 8×8 to 4×4 .
- **Fully Connected Layers:**
 - 256 neurons in the input (obtained from 16 channels of size 4×4).
 - Sequence of layers: $256 \rightarrow 120 \rightarrow 84 \rightarrow 62$, with intermediate ReLU activations.

In this topology, each filter is a learned spline function, making the filters nonlinear and adaptive.

2.3 Topology Comparability

The two architectures, LeNet5 and ConvKAN, are comparable in overall structure while differing in architectural details:

- **General Structure:**
 - Both networks consist of convolutional layers for feature extraction, pooling layers for spatial dimensionality reduction, and fully connected layers for final classification.
- **Type of Convolutional Filters:**
 - **LeNet5:**
 - * Uses linear convolutional filters with fixed scalar weights.
 - * The linearity limits the ability to capture complex nonlinear patterns.
 - **ConvKAN:**
 - * Uses filters based on univariate splines, capable of modeling nonlinear relationships.
- **Weight Parameterization:**
 - **LeNet5:** Weights are scalar parameters learned via backpropagation, with a rigid structure.
 - **ConvKAN:** Weights are represented by spline functions, controlled by parameters such as `grid_size` and `spline_order`, offering a flexible and continuous representation.
- **Number of Neurons in the Fully Connected Layers:**
 - Both networks have 256 neurons in the first fully connected layer, ensuring comparability in the final classification stage.

Implications of the Differences

The architectural differences have significant implications:

- **Performance:**

- ConvKAN may outperform LeNet5 on complex datasets thanks to its ability to capture nonlinear patterns.
- However, the increased computational complexity might lead to overfitting if not properly regularized.

- **Interpretability:**

- LeNet5 offers simpler and more stable activation maps, facilitating interpretation.
- ConvKAN, although more flexible, may produce more complex and less focused activation maps, requiring deeper analysis.

Chapter 3

Dataset and Hyperparameters

3.1 E-MNIST Dataset

- 62 classes (digits and letters).
- Images of size 28×28 in grayscale.
- Split into training, validation (10% extracted from training) and test sets.
- Normalization with mean=0.1307 and std=0.3081.

(Cohen et al., 2017 state: “EMNIST extends MNIST to handwritten letters, providing a more challenging benchmark” [2]).

3.2 Hyperparameters

The hyperparameters were chosen based on previous research on LeNet300 networks and their KAN equivalents to ensure a fair comparison:

- **Learning Rate:** 0.01.
- **Optimizer:** SGD without momentum.
- **Batch Size:** 100.
- **Epochs:** 50.
- **Loss Function:** Cross Entropy.
- **Regularization:** L2 or none.
- **For ConvKAN:** grid_size=5, spline_order=3.

Chapter 4

Grad-CAM: Implementation and Adaptations

4.1 Description of Grad-CAM

Grad-CAM produces activation maps by weighting the feature maps of a convolutional layer with the gradients of the target class. In this way, the areas of the image on which the model focuses are visually highlighted. (Selvaraju et al., 2017 explain: “Grad-CAM produces visual explanations from deep networks via gradient-based localization” [3]).

4.2 Adaptations for ConvKAN

To apply Grad-CAM to ConvKAN, the following adaptations were necessary:

- Identify the final convolutional layer to be used for computing the activation maps.
- Configure hooks to correctly capture both the feature maps and the gradients corresponding to the spline filters.

Chapter 5

Results

5.1 Network Performance

The LeNet5 and ConvKAN networks achieved performances between 90% and 93% on the E-MNIST dataset. In particular, LeNet5 in “No Norm” mode reached approximately 91%, while the use of L2 regularization in ConvKAN allowed for a slight improvement. However, this increment is marginal and comes with increased computational complexity.

5.2 Performance Analysis

To evaluate the training process of LeNet5 and ConvKAN, we analyze the loss and accuracy trends over 50 epochs. The following graphs illustrate these trends for both architectures, considering different normalization settings.

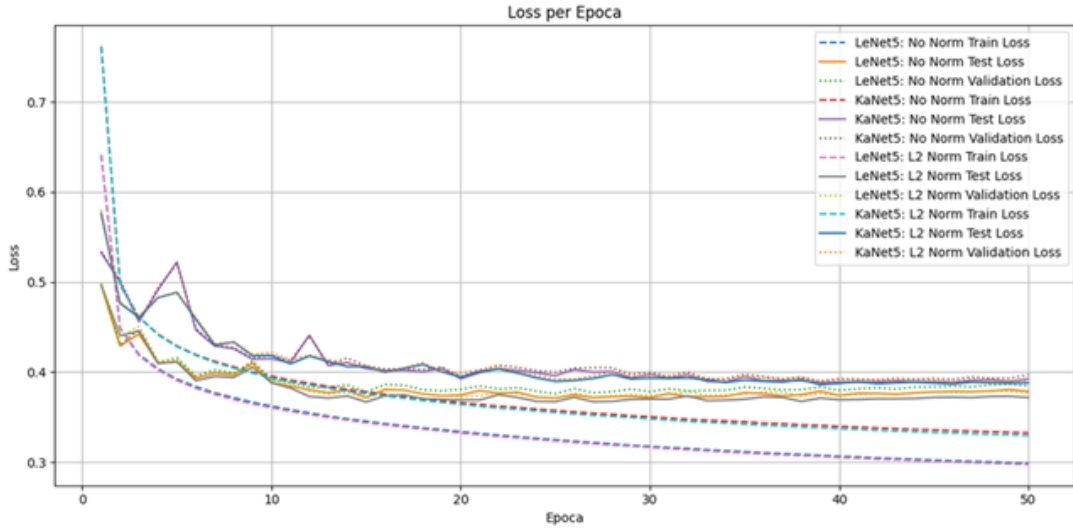


Figure 5.1: Loss trend over epochs for LeNet5 and ConvKAN under different normalization settings.

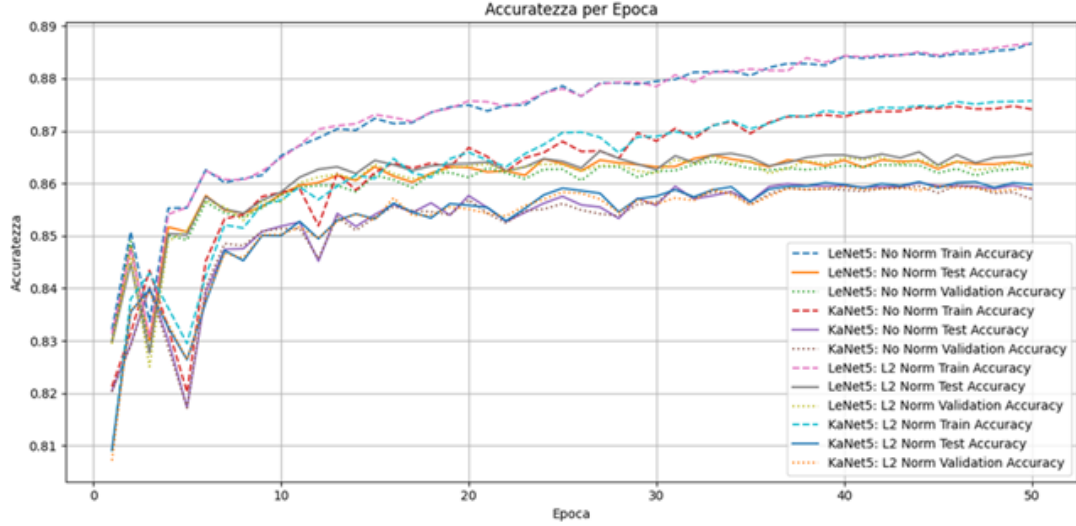


Figure 5.2: Accuracy trend over epochs for LeNet5 and ConvKAN under different normalization settings.

5.3 Visualizations of Heatmaps (Grad-CAM)

- **Correct Examples:** Both networks show focused activation maps. LeNet5 highlights the essential features of the character, while ConvKAN concentrates on the main strokes, albeit with more diffuse activations.
- **Incorrect Examples:** LeNet5's maps appear less sharp yet coherent, whereas ConvKAN's maps are more sparse, reflecting higher internal complexity.
- **Examples where one network fails and the other succeeds:** Differences in attention distribution are evident; ConvKAN may detect patterns that LeNet5 tends to ignore (or vice versa).

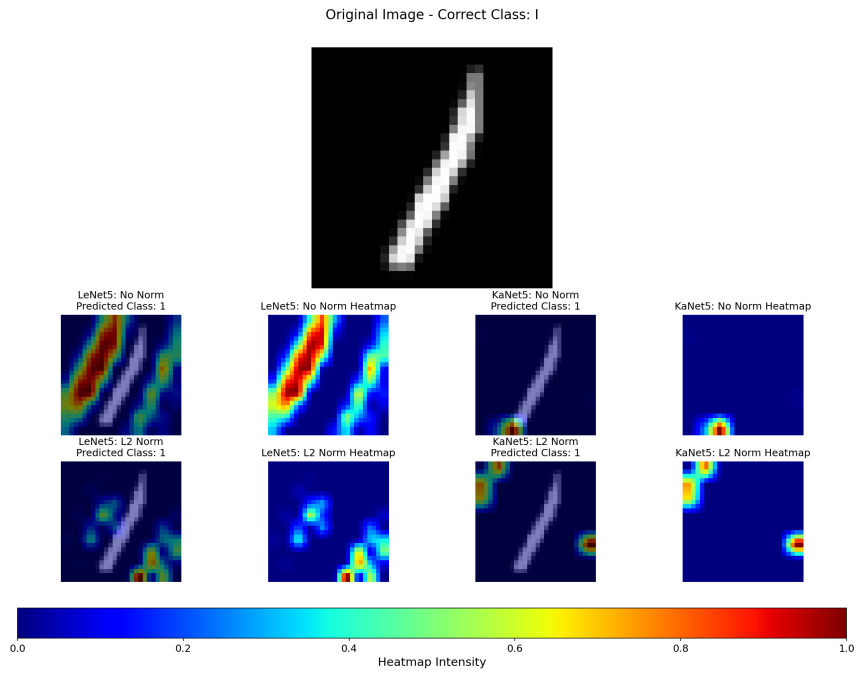


Figure 5.3: Grad-CAM visualization for the character "l".

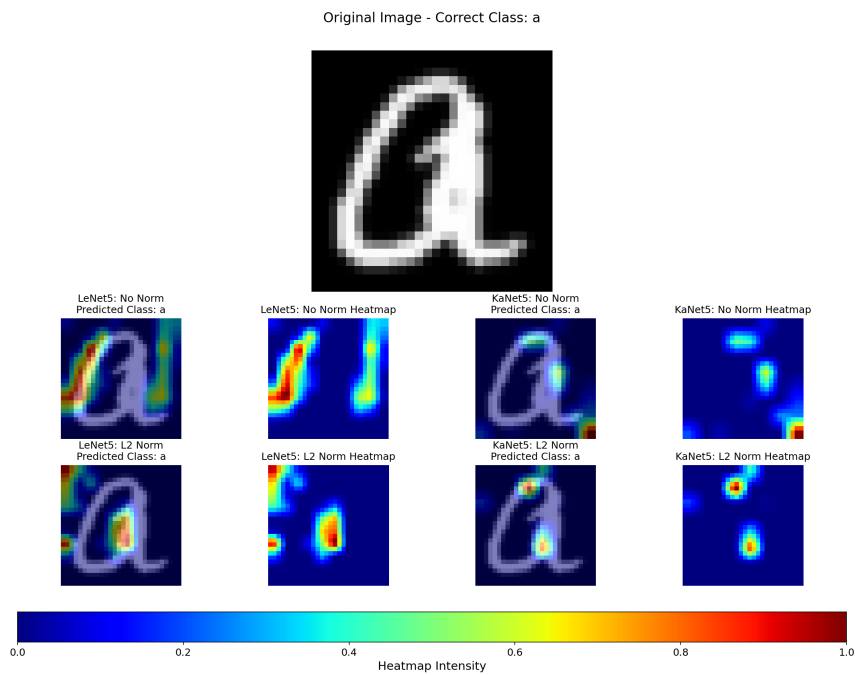


Figure 5.4: Grad-CAM visualization for the character "a".

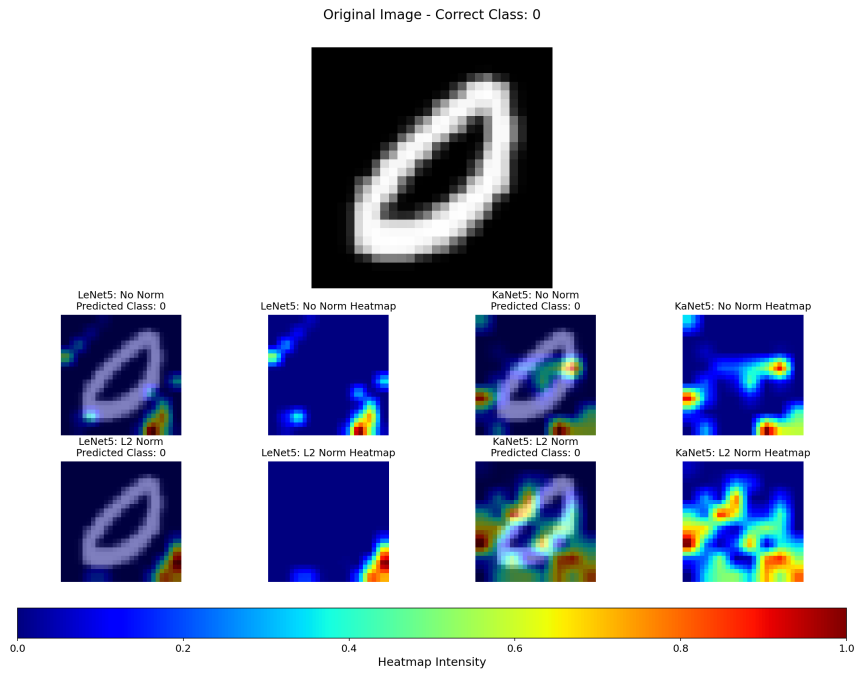


Figure 5.5: Grad-CAM visualization for the digit "0".

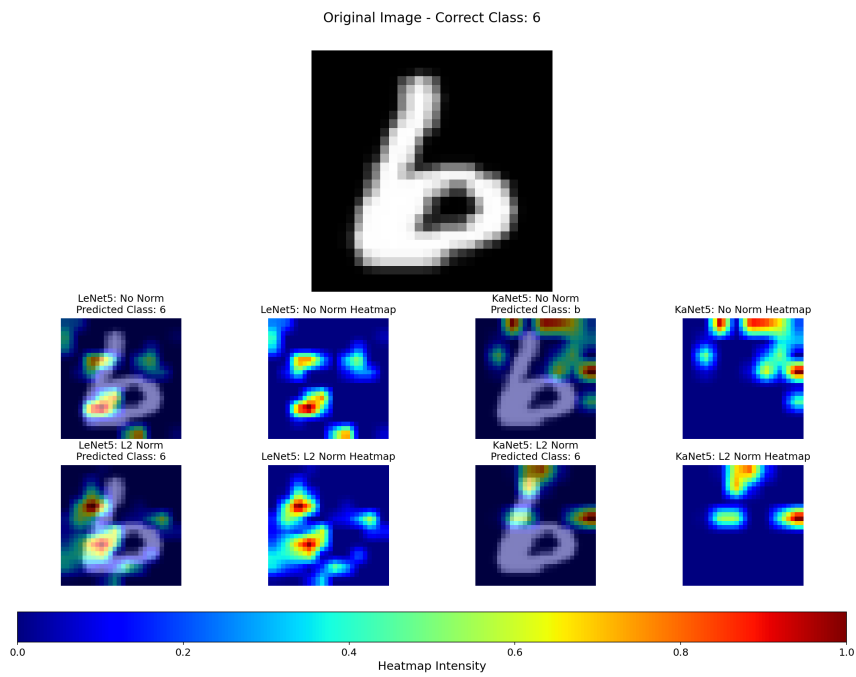


Figure 5.6: Grad-CAM visualization for the digit "6".

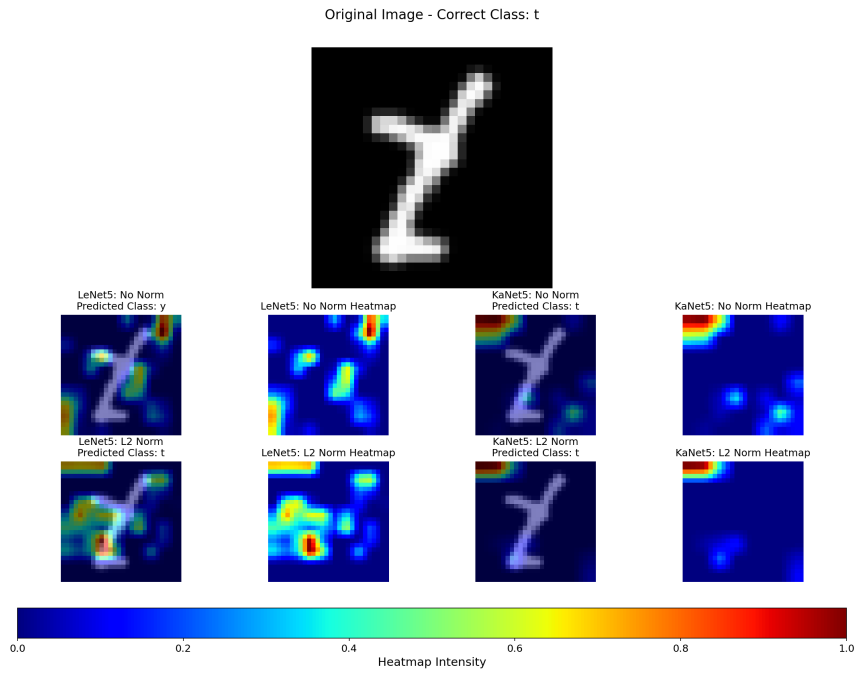


Figure 5.7: Grad-CAM visualization for the character "t".

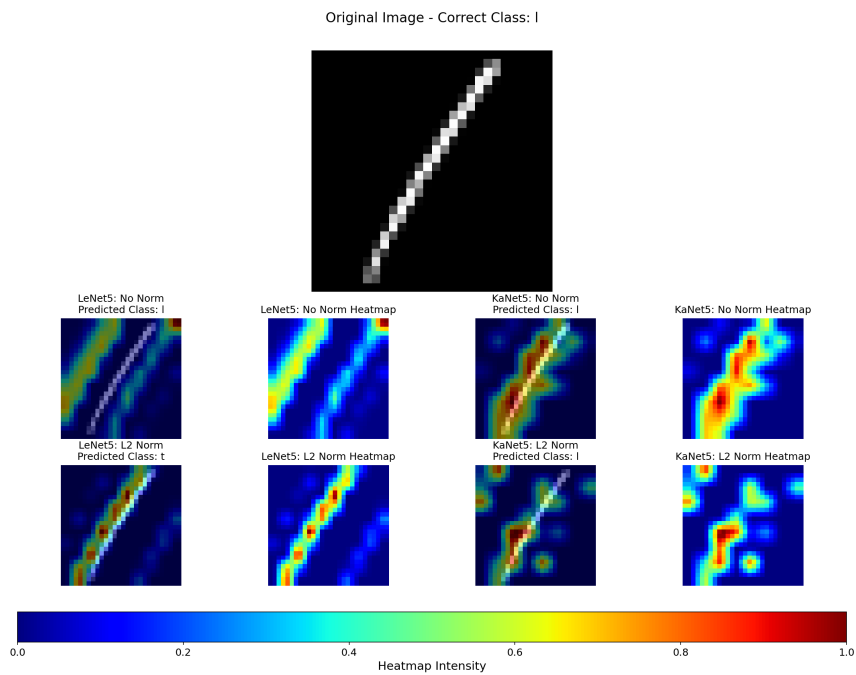


Figure 5.8: Grad-CAM visualization for another instance of the character "l".

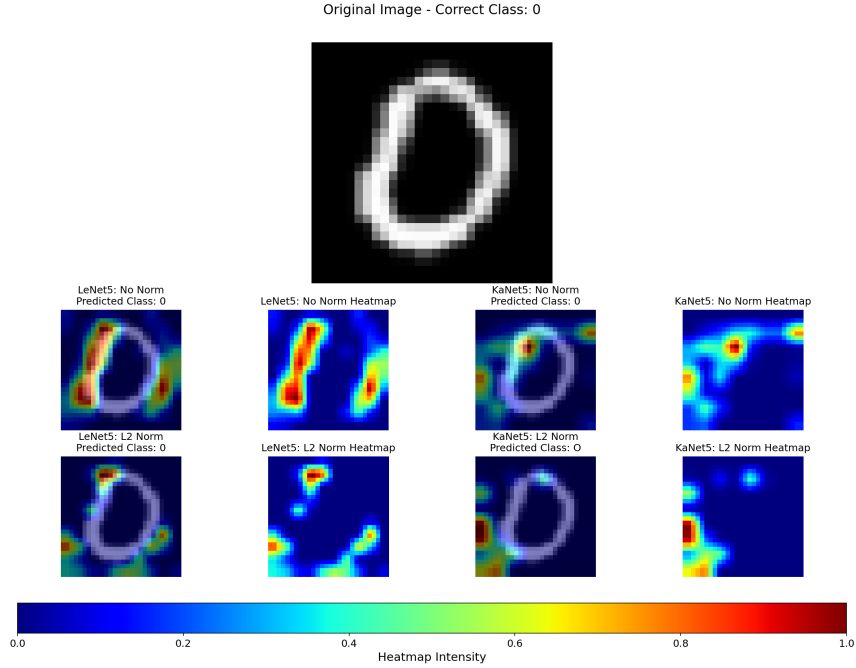


Figure 5.9: Grad-CAM visualization for another instance of the digit "0".

5.4 Heatmap Statistics

The CSV files contain the mean, variance, and maximum values of the Grad-CAM intensity. In correctly classified cases, the intensity is more concentrated, whereas in misclassified cases it decreases. ConvKAN generally shows a higher variance in activations.

Table 5.1: GradCAM Heatmap statistics for LeNet5 in No Norm mode

Description	Mean Intensity	Variance	Max Intensity
Train correct	5.44	167.73	91.79
Train incorrect	5.33	156.76	85.21
Test correct	5.44	167.76	91.81
Test incorrect	5.35	158.63	86.20

Table 5.2: GradCAM Heatmap statistics for KaNet5 in No Norm mode

Description	Mean Intensity	Variance	Max Intensity
Train correct	0.54	427.62	70.93
Train incorrect	0.56	367.81	62.29
Test correct	0.54	427.43	70.95
Test incorrect	0.56	374.89	63.16

Table 5.3: GradCAM Heatmap statistics for LeNet5 in L2 Norm mode

Description	Mean Intensity	Variance	Max Intensity
Train correct	6.34	219.08	103.02
Train incorrect	6.16	201.74	95.14
Test correct	6.34	219.13	103.08
Test incorrect	6.17	204.01	96.15

Table 5.4: GradCAM Heatmap statistics for KaNet5 in L2 Norm mode

Description	Mean Intensity	Variance	Max Intensity
Train correct	0.91	576.16	81.58
Train incorrect	0.82	493.62	71.46
Test correct	0.92	575.79	81.59
Test incorrect	0.81	503.85	72.54

Chapter 6

Extraction of the Mean Example

6.1 Generation and Meaning of the Mean Image

The mean image represents a visual summary of the recurring characteristics in each class of the E-MNIST dataset. In this study, the mean image was obtained by computing the arithmetic average of each pixel across all images belonging to a specific class. The result is a representation that captures the typical shape of the character, highlighting regions where writing features consistently concentrate.

This approach is particularly useful in OCR applications and Explainable AI (xAI), as it provides an objective reference for model attention analysis. By calculating the mean separately for each class, a set of representative images for both letters and digits was created and then used as a base for overlaying the activation maps generated by the convolutional networks. This procedure facilitated a direct comparison between LeNet5 and ConvKAN, offering a clear view of the most activated areas.



Figure 6.1: Mean heatmaps for each class in the E-MNIST dataset.

6.2 Role of the Mean Example Analysis

The mean image played an essential role in model interpretability analysis. It was used as a reference for overlaying activation maps—both those derived from feature maps and those from Grad-CAM—thus facilitating the interpretation of the areas of attention. In particular, by overlaying the mean image, it was possible to evaluate whether the activated regions corresponded to the expected features of the character or if unexpected sub-regions emerged.

6.3 Calculation of the Mean Feature Maps

6.3.1 Extraction and Normalization Method

To obtain the mean feature maps, the activations produced by the last convolutional layer of LeNet5 and ConvKAN were aggregated over a set of test images. For each model, the convolutional output was averaged pixel-by-pixel over all images of the same class, generating a mean map that stably summarizes the model’s response.

A crucial aspect of this analysis is handling the different activation scales between the two models. For this reason, two visualization modes were considered:

1. **Raw:** The feature maps are used with their original values, maintaining the natural scale of the activations.
2. **Normalized:** Each map is scaled to the range $[0,1]$ by dividing by its maximum value, allowing for a qualitative comparison between LeNet5 and ConvKAN.

This dual approach highlights both the absolute differences in activations (using raw maps) and the relative, distributional differences (using normalized maps). The procedure also included resampling the feature maps—often of lower resolutions (e.g., 8×8 or 5×5)—using bilinear interpolation to align them with the mean image (28×28), ensuring coherent overlay for visual analysis.

6.3.2 Interpretation of the Feature Maps

The analysis of the mean feature maps revealed significant differences in how the two models interpret the inputs. LeNet5 produces relatively uniform and continuous activations covering the entire characteristic area in a diffused manner, highlighting both the contours and inner regions of the character. This behavior indicates a global and coherent response typical of linear convolutional filters.

6.4 Calculation of the Mean GradCAM Maps

6.4.1 GradCAM Implementation: Parameters and Adaptations for ConvKAN

The GradCAM technique (Gradient-weighted Class Activation Mapping) was adapted for both LeNet5 and ConvKAN to highlight the decisive areas for classification. During the forward pass, the output of the last convolutional layer is saved; during the backward

pass, the gradients with respect to the target class are recorded. These gradients are averaged per channel and combined with the corresponding feature maps to produce a heatmap that highlights the regions of greatest importance.

In the case of ConvKAN, due to the spline nature of its filters, custom callbacks were implemented to ensure that the gradients were correctly computed and assigned to each filter. The GradCAM parameters, such as the gradient aggregation method and the weighting mechanism of the feature maps, were optimized to ensure that the resulting heatmap faithfully represents the decisive areas for classification in both architectures.

6.4.2 Differences in Activation between GradCAM and Feature Maps

While the mean feature maps provide an overview of the raw activations from the convolutional layer, GradCAM highlights the regions that have a decisive impact on the final output. In other words, the feature maps indicate *where* the model activates, whereas GradCAM quantifies *how much* each activation contributes to the classification decision.

In our study, LeNet5 tends to produce GradCAM maps that overlap homogeneously across the entire structure of the character, reflecting the global behavior of its linear filters. In contrast, ConvKAN shows more segmented and localized GradCAM maps, consistent with its spline-based architecture.

6.5 New Metrics for Advanced Analysis

The new metrics introduced—Loop Fragmentation, Vertical Continuity Ratio, and Open Letter Ratio—provide quantitative tools to characterize the distribution and segmentation of activations, highlighting structural differences between LeNet5 and ConvKAN.

6.5.1 Loop Fragmentation

The Loop Fragmentation metric quantifies the degree of segmentation in the activated regions corresponding to the internal loops of letters. Essentially, this metric provides a count of how many distinct, non-connected regions exist within the area of a loop, indicating the extent to which the loop is fragmented in the activation map.

To compute this metric, we follow a series of well-defined steps:

1. **Normalization:** We start with the raw activation heatmap and normalize it by dividing each pixel value by the maximum observed value, producing a normalized heatmap

$$H : \Omega \subset \mathbb{R}^2 \rightarrow [0, 1],$$

where $H(x, y)$ represents the normalized activation at pixel (x, y) .

2. **Thresholding:** A fixed threshold τ (typically set to $\tau = 0.3$) is applied to the normalized heatmap to produce a binarized version $H_n : \Omega \rightarrow \{0, 1\}$:

$$H_n(x, y) = \begin{cases} 1, & \text{if } H(x, y) \geq \tau, \\ 0, & \text{if } H(x, y) < \tau. \end{cases}$$

3. **Connected Component Labeling (CCL):** With the binarized heatmap, contiguous regions are identified and labeled using a two-pass connected component labeling algorithm. The total number of distinct connected components k represents the degree of fragmentation.
4. **Metric Definition:** The Loop Fragmentation metric (LF) is defined as:

$$LF = k.$$

A value of $LF = 1$ indicates a continuous, unfragmented loop, whereas $LF > 1$ signifies that the loop is segmented into multiple fragments.

6.5.2 Vertical Continuity Ratio

This metric measures the continuity of activations along the vertical axis of the heatmap, which is particularly relevant for letters with vertical strokes (e.g., “I”, “l”, “1”). It essentially evaluates whether the activations remain uniform along the height of the letter or if there is a notable drop in the central region.

To compute the Vertical Continuity Ratio (VCR), we proceed as follows:

1. **Row-wise Maximum Extraction:** For each row r of the normalized heatmap H , we compute the maximum activation value across all columns:

$$p(r) = \max_{c \in \{0, \dots, W-1\}} H(r, c),$$

where W is the width of the heatmap.

2. **Vertical Segmentation and Averaging:** The heatmap is divided into three vertical segments: top, middle, and bottom. Let μ_{middle} be the average of the maximum values in the middle segment and μ_{TB} be the average over the combined top and bottom segments.
3. **VCR Calculation:** The Vertical Continuity Ratio is defined as:

$$VCR = \frac{\mu_{\text{middle}}}{\mu_{\text{TB}} + \varepsilon},$$

where ε is a small constant to prevent division by zero.

6.5.3 Open Letter Ratio

The Open Letter Ratio (OLR) quantifies the proportion of activation that falls within the “open” areas of a letter. This metric is computed only for specific classes—namely, those corresponding to letters with open interiors (e.g., C, G, J, S, Z). Its purpose is to assess whether the network’s attention extends into the internal void of a letter or is concentrated predominantly along its contours.

To compute the OLR, we adopt the following procedure:

1. **Mean Image Thresholding and Bounding Box Extraction:** Starting from the mean image of the class (obtained by averaging all images belonging to that class), a threshold (for example, via Otsu’s method) is applied to produce a binary image that highlights the character. The minimal bounding box that encloses all foreground pixels is then extracted.

2. **Loading the Open Mask:** For the designated open classes, a hand-crafted mask is used in which the open areas are marked in red.
3. **Mask Resizing:** Both the open mask and the binary letter mask (from the thresholded mean image) are resized to match the dimensions of the activation heatmap using nearest-neighbor interpolation, ensuring proper alignment.
4. **Ratio Calculation:** Finally, the OLR is computed on the activation heatmap $H(x, y)$ by summing the activation values within the open mask region and dividing by the sum of activation values over the entire letter region:

$$OLR = \frac{\sum_{(x,y) \in \text{open mask}} H(x, y)}{\sum_{(x,y) \in \text{letter mask}} H(x, y) + \varepsilon},$$

where ε is a small constant to avoid division by zero.

Chapter 7

Dynamic Visualization System

The web interface, developed using HTML, CSS, and JavaScript, enables interactive exploration of the evolution of activation maps over time. Users can enter an E-MNIST character string into a text input field, dynamically selecting which images to juxtapose for comparison. The displayed images update in real-time, showing how heatmaps evolve across training epochs.

To enhance user experience and facilitate analysis, the interface includes several key features:

- **Auto-loop:** Each video automatically restarts after a short pause at the end of the animation.
- **Playback Speed Control:** A slider allows users to modify the playback speed nonlinearly, making it possible to observe both rapid and slow transitions.
- **Navigation and Zoom:** The interface supports pan and zoom functionalities, enabling users to closely examine specific regions of the activation maps.
- **Auto-sizing:** The visualization automatically adjusts to different screen sizes and image dimensions, optimizing the layout for clarity and ease of use.

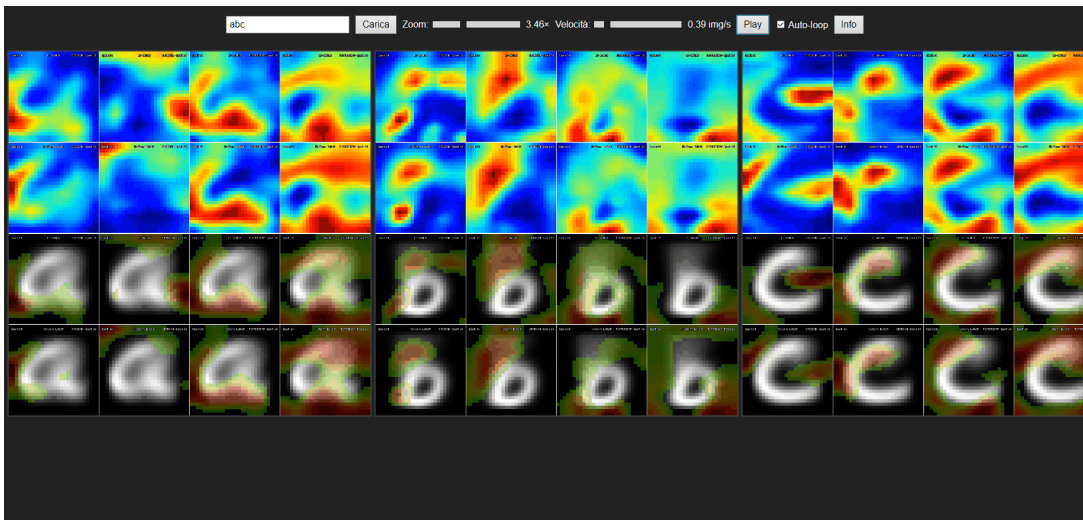


Figure 7.1: Web interface for dynamic visualization of activation maps. Users can select characters to compare, observe heatmap evolution across epochs, and navigate images using pan and zoom functionalities.

Chapter 8

Statistical Analysis and Comparisons

Statistical analysis compared Standard (LeNet5) and ConvKAN (KaNet5) across various metrics, evaluating differences between Feature Maps and GradCAM in both raw and normalized versions.

8.0.1 Corner and Blur

The *corner* and *blur* metrics provide structural information about the activations and their interpretability:

1. **Corner Score:** Measures the density of corner points in the activations, computed via skeletonization of the mean image and corner detection using the Harris method. A high score indicates greater sensitivity to sharp directional changes, particularly relevant for letters with angular strokes.
2. **Blur Score:** Quantifies the blurriness of the activations by measuring the local variance of intensity gradients. A high value suggests diffuse, less focused activations, while a low value indicates sharp, concentrated activations.

T-tests were performed to compare these metrics between Standard and ConvKAN, analyzing both Feature Maps and GradCAM maps.

8.0.2 Analysis of Additional Metrics

In addition to *corner* and *blur*, further metrics were introduced:

1. **Loop Fragmentation:** Measures the fragmentation of activations in the loops of letters (e.g., “O”, “B”, “D”).
2. **Vertical Continuity Ratio:** Evaluates the consistency of activation along the vertical axis in letters with strokes (e.g., “I”, “l”, “1”).
3. **Open Letter Ratio:** Quantifies the activation in the open areas of letters (e.g., “C”, “G”), checking whether the model recognizes the empty space as part of the character structure.

Chapter 9

Results and Interpretations in the xAI Context

The aggregated results reveal contrasting patterns between Feature Maps and GradCAM. For Feature Maps (Raw), Standard (LeNet5) exhibits markedly higher overall activations (5.94 ± 0.69 vs. 0.55 ± 0.72), corner activations (5.69 ± 0.66 vs. 0.54 ± 0.69), blur activations (6.26 ± 0.68 vs. 1.80 ± 0.60), and non-blurred activations (5.75 ± 0.75 vs. 2.03 ± 0.50) compared to ConvKAN (KaNet5). In contrast, for GradCAM (Raw) the trend is inverted, with ConvKAN showing higher overall (0.17 ± 0.06 vs. 0.11 ± 0.02), corner (0.16 ± 0.06 vs. 0.11 ± 0.02), blur (0.16 ± 0.07 vs. 0.12 ± 0.03), and non-blurred activations (0.16 ± 0.06 vs. 0.11 ± 0.02).

A similar pattern is observed in the normalized domain: Feature Maps from Standard exhibit higher values (overall: 0.62 ± 0.05 , corner: 0.59 ± 0.04 , blur: 0.65 ± 0.06 , non-blurred: 0.60 ± 0.05) than those from ConvKAN (overall: 0.10 ± 0.15 , corner: 0.10 ± 0.15 , blur: 0.35 ± 0.11 , non-blurred: 0.39 ± 0.06), whereas for GradCAM, ConvKAN produces higher activations (overall: 0.45 ± 0.11 vs. 0.36 ± 0.07 , corner: 0.43 ± 0.11 vs. 0.34 ± 0.06 , blur: 0.45 ± 0.13 vs. 0.39 ± 0.08 , non-blurred: 0.45 ± 0.11 vs. 0.34 ± 0.07).

Table 9.1: Aggregated FeatureMap (Raw)

Metric	LeNet5	KaNet5
overall_activation	5.94 ± 0.69	0.55 ± 0.72
corner_activation	5.69 ± 0.66	0.54 ± 0.69
blur_activation	6.26 ± 0.68	1.80 ± 0.60
non_blurred_activation	5.75 ± 0.75	2.03 ± 0.50

Table 9.2: Aggregated GradCAM (Raw)

Metric	LeNet5	KaNet5
overall_activation	0.11 ± 0.02	0.17 ± 0.06
corner_activation	0.11 ± 0.02	0.16 ± 0.06
blur_activation	0.12 ± 0.03	0.16 ± 0.07
non_blurred_activation	0.11 ± 0.02	0.16 ± 0.06

Table 9.3: Aggregated FeatureMap (Normalized)

Metric	LeNet5	KaNet5
overall_activation	0.62 ± 0.05	0.10 ± 0.15
corner_activation	0.59 ± 0.04	0.10 ± 0.15
blur_activation	0.65 ± 0.06	0.35 ± 0.11
non_blurred_activation	0.60 ± 0.05	0.39 ± 0.06

Table 9.4: Aggregated GradCAM (Normalized)

Metric	LeNet5	KaNet5
overall_activation	0.36 ± 0.07	0.45 ± 0.11
corner_activation	0.34 ± 0.06	0.43 ± 0.11
blur_activation	0.39 ± 0.08	0.45 ± 0.13
non_blurred_activation	0.34 ± 0.07	0.45 ± 0.11

9.1 Interpretation of Mean Heatmaps: Feature Maps and GradCAM

Feature Maps:

1. **Corner Activation:** In normalized mode, Standard exhibits a mean corner activation of approximately 0.59, whereas ConvKAN shows a much lower value of around 0.10.
2. **Blur Activation:** Standard reaches a normalized mean blur activation of about 0.65, compared to 0.35 for ConvKAN.

These observations suggest that Standard emphasizes structural features with more globally distributed activations in its feature maps. In contrast, ConvKAN produces considerably lower activations in the same modality.

GradCAM:

1. **Corner Activation:** For GradCAM maps, ConvKAN yields a higher normalized corner activation (approximately 0.43) than Standard (around 0.34), indicating a more focused concentration on key regions.
2. **Blur Activation:** Similarly, ConvKAN’s GradCAM maps present higher blur activations (about 0.45) compared to Standard (approximately 0.39).

9.2 Interpretation of the New Metrics

The new metrics — loop fragmentation, vertical continuity ratio, and open letter ratio — further elucidate the internal behavior of the models.

1. **Loop Fragmentation:** In normalized mode using Feature Maps, ConvKAN fragments internal loops more strongly than Standard (mean: 2.43 vs. 1.00, $p < 0.01$). Conversely, for GradCAM maps, Standard exhibits higher loop fragmentation (mean: 1.82 vs. 1.37, $p < 0.01$).
2. **Vertical Continuity Ratio:** For Feature Maps, Standard shows a higher vertical continuity ratio (mean: 1.22) than ConvKAN (mean: 0.94), although this difference is marginally non-significant ($p = 0.07$). In contrast, for GradCAM maps, ConvKAN demonstrates a significantly higher ratio (1.42 vs. 0.93, $p = 0.02$).
3. **Open Letter Ratio (OLR):** When computed on the designated open classes, the OLR derived from Feature Maps is similar between Standard (mean: 0.31) and ConvKAN (mean: 0.30, $p = 0.43$). However, for GradCAM maps, Standard shows a significantly higher OLR (mean: 0.32) compared to ConvKAN (mean: 0.27, $p = 0.03$).

Table 9.5: New Metrics Comparison (Normalized) for Feature Maps and GradCAM

Metric*	Modality	Standard (LeNet5)	ConvKAN (KaNet5)
Loop Fragmentation (LF*)	Feature Maps	1.00	2.43 ($p < 0.01$)
	GradCAM	1.82	1.37 ($p < 0.01$)
Vertical Continuity Ratio (VCR*)	Feature Maps	1.22	0.94 ($p = 0.07$)
	GradCAM	0.93	1.42 ($p = 0.02$)
Open Letter Ratio (OLR*)	Feature Maps	0.31	0.30 ($p = 0.43$)
	GradCAM	0.32	0.27 ($p = 0.03$)

Table 2 Description: The metrics marked with an asterisk (*) are evaluated under the null hypotheses that there are no significant differences between the two models for the given metric, tested separately for both the Feature Maps and GradCAM modalities.

9.3 Summary and Conclusions

The experimental results reveal a contrasting behavior between the two modalities. For Feature Maps, Standard (LeNet5) consistently produces higher activations across all measures (overall, corner, blur, and non-blurred) compared to ConvKAN, indicating a more globally integrated representation of character features. In contrast, for GradCAM, ConvKAN (KaNet5) yields higher activations, suggesting a more focused localization of salient regions.

Regarding the new metrics, the data indicate that in normalized mode:

- **Loop Fragmentation:** ConvKAN exhibits more fragmented loop activations in Feature Maps, while Standard shows higher fragmentation in GradCAM.
- **Vertical Continuity Ratio:** Standard achieves a higher ratio in Feature Maps (implying more continuous vertical structures), whereas ConvKAN records a significantly higher ratio in GradCAM.
- **Open Letter Ratio:** The OLR is comparable between the models in Feature Maps; however, in GradCAM, Standard allocates more attention to the open areas of letters.

Conclusions:

The results indicate that Standard (LeNet5) produces more globally distributed and higher-magnitude activations in its Feature Maps, while ConvKAN (KaNet5) emphasizes localized activations in its GradCAM outputs. These differences are reflected in both the primary activation metrics and the additional measures of loop fragmentation, vertical continuity, and open letter ratio. Overall, the data provide insights into how each model allocates attention to various structural features in handwritten characters.

9.4 Implications for Explainability

Overall, the findings suggest that Standard (LeNet5) offers a comprehensive representation of the characters via its high and uniformly distributed activations in Feature Maps, which may facilitate global interpretation. Conversely, ConvKAN (KaNet5) produces more focused and localized activations in GradCAM, which could enhance interpretability by pinpointing key regions. This concentrated focus may prove particularly advantageous in tasks involving images with multiple objects, where fine-grained discrimination among observed objects is essential. In such cases, the ability of ConvKAN to isolate and emphasize salient regions could improve both explainability and performance in tasks such as object detection or segmentation—even though this might come at the cost of a less cohesive overall activation pattern.

9.5 Future Directions and Research Opportunities

Our study builds on a growing body of work exploring the potential of KAN-based architectures. In recent years, several new papers have demonstrated the advantages of such architectures in various domains. For example, Li et al. [6] showed that KAN-based models can serve as a strong backbone for medical image segmentation and generation, suggesting that architectures with pinpointing capabilities may excel in multi-object scenarios. Similarly, Jamali et al. [7] explored KANs for hyperspectral image classification, highlighting their ability to capture subtle, localized features in high-dimensional data.

- **Expanding to Diverse Datasets:** Extending the analysis to larger and more diverse datasets—such as ImageNet, COCO, or other domain-specific collections—could verify the generalizability of the observed activation patterns and test the robustness of both architectures under varied conditions. It is also worth noting that the use of EMNIST, which contains only a single object per image, might have limited the advantages of ConvKAN’s pinpointing capability. Evaluating datasets with multiple objects could better highlight ConvKAN’s strength in localizing features.
- **Exploration of New Vision Tasks:** While the present work focuses on handwritten character recognition, applying similar interpretability and activation analysis methods to other computer vision tasks—such as object detection, semantic segmentation, and scene understanding—could broaden our understanding of model behavior. For instance, the promising results obtained by Ma et al. [8] in remote sensing image semantic segmentation and by Cambrin et al. [9] in crop field segmentation suggest that focused, localized activations may enhance performance in complex, multi-object environments.

- **Integration of Hybrid Architectures:** The pinpointing capability demonstrated by ConvKAN, as evidenced by its more focused GradCAM activations, suggests that integrating attention mechanisms or combining traditional CNNs with emerging architectures (such as CNN-Transformer hybrids) might further enhance both performance and interpretability in complex vision tasks.
- **Advanced Interpretability Techniques:** Future work could benefit from combining the current metrics (Loop Fragmentation, Vertical Continuity Ratio, and Open Letter Ratio) with additional advanced interpretability techniques—such as multi-layer saliency mapping, feature inversion, or localized relevance propagation—to develop a more comprehensive understanding of model decision processes.
- **Segmentation Applications:** The precise pinpointing ability of ConvKAN may translate into a performance advantage in tasks that require accurate localization of features, such as medical image segmentation or satellite image segmentation. Investigating whether the focused activation patterns of ConvKAN can improve boundary detection and overall segmentation accuracy in these contexts would be a valuable extension of this work. In this regard, FloodKAN: Integrating Kolmogorov–Arnold Networks for Efficient Flood Extent Extraction by Wang et al. [10] has demonstrated significant improvements in flood segmentation accuracy using a KAN-based model, underscoring the potential benefits of such architectures for multi-object segmentation tasks in other domains.

These future research directions aim not only to enhance model performance but also to deepen our understanding of the internal mechanisms driving interpretability in computer vision applications.

Bibliography

- [1] LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). *Gradient-Based Learning Applied to Document Recognition*. Proceedings of the IEEE, 86(11), 2278-2324.
- [2] Cohen, G., Afshar, S., Tapson, J., & van Schaik, A. (2017). *EMNIST: Extending MNIST to handwritten letters*. arXiv preprint arXiv:1702.05373.
- [3] Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). *Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization*. Proceedings of the IEEE International Conference on Computer Vision (ICCV).
- [4] Liu, Z., Wang, Y., Vaidya, S., Ruehle, F., Halverson, J., Soljačić, M., Hou, T. Y., & Tegmark, M. (2024). *Kolmogorov-Arnold Networks (KANs): Promising alternatives to Multi-Layer Perceptrons (MLPs)*. Preprint. Retrieved from <https://arxiv.org/abs/2404.19756>.
- [5] Bodner, A. D., Tepsich, A. S., Spolski, J. N., & Pourteau, S. (2024). *Convolutional Kolmogorov-Arnold Networks*. arXiv preprint arXiv:2406.13155. <https://doi.org/10.48550/arXiv.2406.13155>.
- [6] Li, C.; Liu, X.; Li, W.; Wang, C.; Liu, H.; Liu, Y.; Chen, Z.; Yuan, Y. (2024). *U-KAN Makes Strong Backbone for Medical Image Segmentation and Generation*. arXiv preprint arXiv:2406.02918.
- [7] Jamali, A.; Roy, S.K.; Hong, D.; Lu, B.; Ghamisi, P. (2024). *How to Learn More? Exploring Kolmogorov-Arnold Networks for Hyperspectral Image Classification*. Remote Sens., 16, 4015.
- [8] Ma, X.; Wang, Z.; Hu, Y.; Zhang, X.; Pun, M.-O. (2025). *Kolmogorov-Arnold Network for Remote Sensing Image Semantic Segmentation*. arXiv preprint arXiv:2501.07390.
- [9] Cambrin, D.R.; Poeta, E.; Pastor, E.; Cerquitelli, T.; Baralis, E.; Garza, P. (2024). *KAN You See It? KANs and Sentinel for Effective and Explainable Crop Field Segmentation*. arXiv preprint arXiv:2408.07040.
- [10] Wang, C.; Zhang, X.; Liu, L. (2025). *FloodKAN: Integrating Kolmogorov-Arnold Networks for Efficient Flood Extent Extraction*. Remote Sens., 17, 564. <https://doi.org/10.3390/rs17040564>.